



The potential of seep carbonates to preserve the seawater isotope compositions



Yinan Deng^{1,2}✉, Qingjun Guo^{3,4}✉, Ganglan Zhang², Fang Chen², Shengxiang Yang², Gaowen He², Hongfei Lai², Daohua Chen², Yunxin Fang¹, Zenggui Kuang¹, Jun Cao², Yangtao Zhu², Dawuda Usman³, Yufei Liu², Bin Zhao², Xuexiao Jiang² & Mingyu Zhao⁵

The isotopic compositions of authigenic minerals of marine sediments are potential proxies for reconstructing paleoenvironmental conditions. However, the reliability and applicability of this tracer remain elusive due to the limited spatial distribution scope and alteration during early diagenesis. Here we report isotopic measurements of seep carbonates (calcium, molybdenum, strontium, neodymium) generated by gas hydrate dissociation, sampled from a core drilled at the South China Sea. Our results show that there is approximately -1.2% calcium isotopic fractionation between aragonite and seawater. On the other hand, the molybdenum, strontium, and neodymium isotopic compositions are similar to those found in modern seawater. These results indicate that aragonite-dominated seep carbonate, which preferentially precipitates in proximity to the seawater-sediment interface, can preserve the pristine seawater isotopic signatures. Therefore, we infer that aragonite-dominant seep carbonate is a promising archive for coeval seawater chemistry.

Changes in seawater chemistry are intertwined with the evolution of the overall Earth system^{1–3}. The isotopic compositions of authigenic sedimentary phases, including bioapatite, Fe–Mn oxides, and authigenic carbonate, are valuable archives for seawater compositions, and have been widely used in paleoenvironment studies^{4–7}. However, tracking the isotopic variations of seawater through time still remains difficult due to the lack of a continuous record⁵. The biological factors, so-called “vital effects”, also alter some isotope systems, limiting their use as oceanic tracers^{8,9}. In addition, the application of the geochemical paleoredox proxies can be limited by our ability to recognize records impacted by diagenesis^{10–12}, as pore water often exhibits geochemical compositions distinct from seawater^{7,13,14}. While developing a universally applicable archive for the authigenic phase from different sediments poses challenges, establishing such a framework is essential to advance the use of Ca–Mo–Sr–Nd isotopic records as robust paleoceanographic tracers.

Methane seeps are widely observed on active and passive continental margins worldwide throughout much of Earth’s history^{15–17}, and are commonly associated with the dissociation of gas hydrates^{18,19}. The sulfate-driven anaerobic oxidation of methane (AOM) has been shown to consume 90% or more of the methane produced in subsurface seafloor

environments²⁰. Seep carbonates, by-products of methane seepage, are formed widely close to the seafloor^{21,22} due to the AOM coupled with sulfate reduction²³. It was suggested that seep carbonates represent excellent archives of past methane seepage and paleoenvironmental conditions^{16,19,24}. A recent study from the South China Sea (SCS) indicated the potential for seep carbonates to preserve the (near) seawater molybdenum isotopic composition, whereas there is still an isotopic offset of $\sim 0.5\%$ from seawater²⁵. The Ca–Mo–Sr–Nd isotopic compositions of seawater hold important information on past environmental changes and marine geochemical cycles. Aragonite, one of the authigenic carbonates in seep settings, preferentially forms near the seawater-sediment interface^{26,27}. Their formation at or near the seafloor also reduces exposure to diagenesis in pore water, which commonly alters redox-sensitive proxies in deeper sediments^{26,27}. Consequently, aragonite and co-precipitated authigenic minerals (e.g., sulfides) at methane seeps may serve as reliable archives for seawater chemistry. Aragonite, or formerly aragonite, is also commonly found in seep settings in geological history^{28,29}. However, studies assessing the reliability of authigenic carbonates in seep settings as archives of coeval seawater isotopic records remain scarce.

¹National Engineering Research Center for Gas Hydrate Exploration and Development, Guangzhou, China. ²Key Laboratory of Marine Mineral Resources, Ministry of Natural Resources, Guangzhou Marine Geological Survey, China Geological Survey, Guangzhou, China. ³Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing, China. ⁴College of Resources and Environment, University of Chinese Academy of Sciences, Beijing, China. ⁵Key Laboratory of Cenozoic Geology and Environment, Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China.

✉ e-mail: dengyan@126.com; guojq@igsrr.ac.cn

Although previous studies have progressively investigated the Mo–Ca–Sr–Nd isotopes of seep carbonates, their focus has predominantly been on the isotopic fractionation mechanisms and compositional characteristics within cold seep systems^{25,30}. The Ca isotopes record carbonate precipitation kinetics modulated by AOM, while Mo isotopes reveal redox transitions (suboxic–anoxic–sulfidic) in cold seep environments. Radiogenic Sr and Nd isotopes effectively delineate the origins of seep fluids (e.g., seawater versus pore water) contributing to carbonate formation. The combined Ca–Mo–Sr–Nd isotopic systems synergistically constrain fluid sources, seepage flux magnitudes, and redox dynamics during carbonate precipitation, offering a multi-proxy framework to decode seep system evolution. This approach resolves ambiguities inherent to conventional proxies (e.g., $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, trace elements), which often provide indirect or conflated constraints on environmental conditions. By resolving Mo–Ca–Sr–Nd isotopic behaviors and linking them to fluid and redox dynamics, which would open avenues for paleoenvironmental studies. The isotopic signatures preserved in seep carbonates, particularly their fidelity to seawater chemistry, provide a baseline for reconstructing past environmental changes and marine geochemical cycles. Here, we present the multiple (i.e., Mo, Ca, Sr and Nd) isotopic values, in combination with other geochemical data, in the different authigenic minerals of 14 layers of seep

carbonate from a drill core recovered during China’s second gas hydrate drilling expedition (GMGS-2) in the SCS (Fig. 1). The present study provides new insight into the isotope systematics of seep carbonates. In addition, it provides an opportunity to evaluate whether these sediments can preserve the pristine seawater signatures.

Geological background

The study area (GMGS2-08 site) is located in the northeastern part of the SCS, which is one of the largest marginal seas in the West Pacific (Fig. 1). This site was drilled to a depth of ~95 meters below the seafloor (mbsf) and is dominated by clayey silt and silt, at water depths of ~800 m with high deposition rate of ~54.2 cm/ka over the past 0.12 Ma time period^{19,24}. Three stages of methane seepage: ~130.3 ka BP before, MIS 5 (~130.3–111.4 ka BP), and MIS 1 (~11.1–10.0 ka BP) have been observed. Massive aragonite-dominant seep carbonates formed under intermittently euxinic conditions with enhanced methane seepage close to or at the sediment–seawater interface during stage 2. The typical textures of well-preserved, needle-like aragonite crystals (Supplementary Fig. 1), coupled with $\delta^{13}\text{C}$ values (–31.6‰ to –56.8‰) and $\delta^{18}\text{O}$ values (average: ~3.64‰) commonly found in methane-rich fluids, confirm that aragonite-dominated carbonates retain primary isotopic compositions. Stages 1 and 3 are dominated by high-

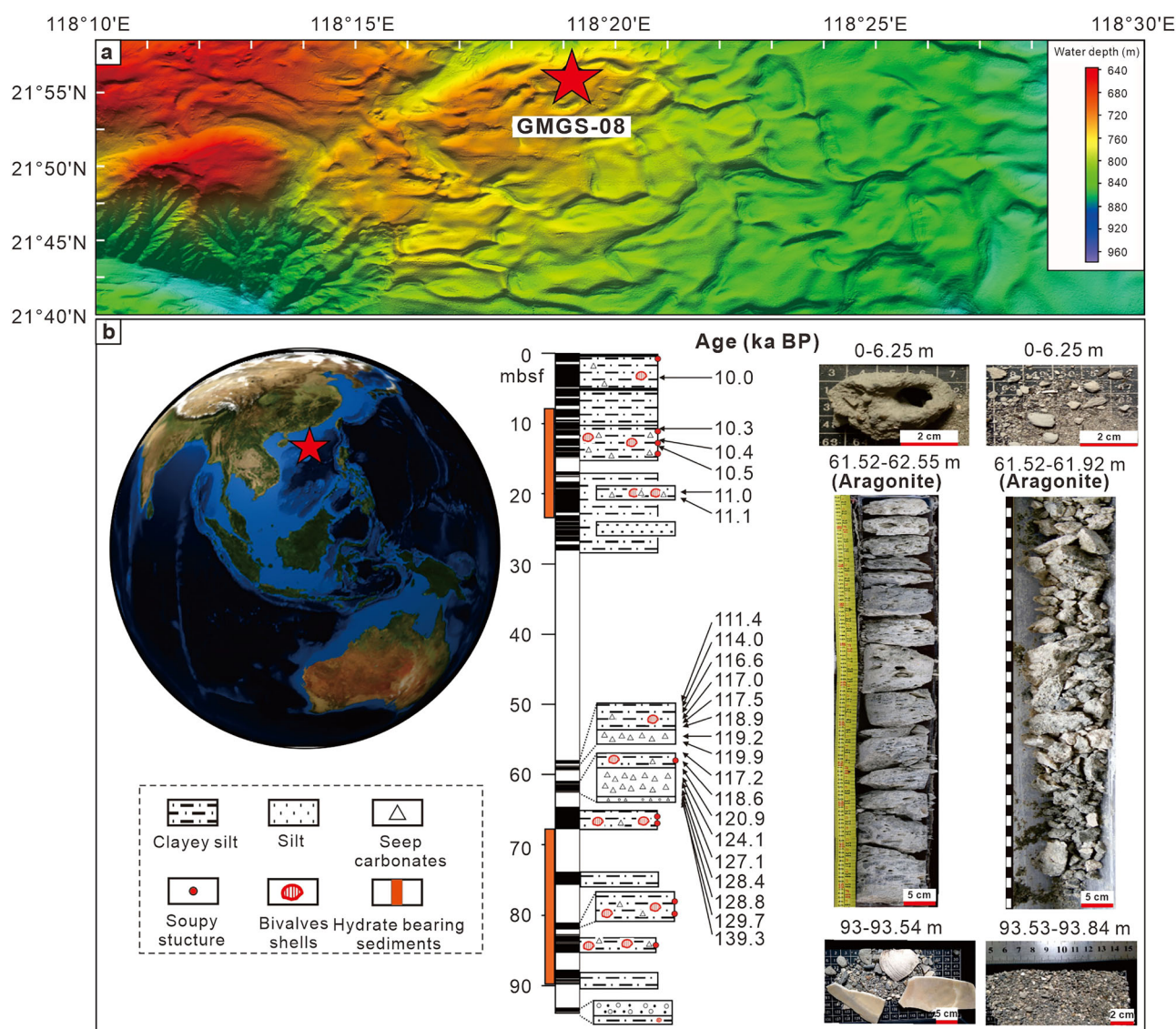


Fig. 1 | Location and lithological columns of the studied samples. **a** High-resolution bathymetry maps of the study area. **b** The location, lithological columns, and distribution of seep carbonates at the GMGS2-08 site.

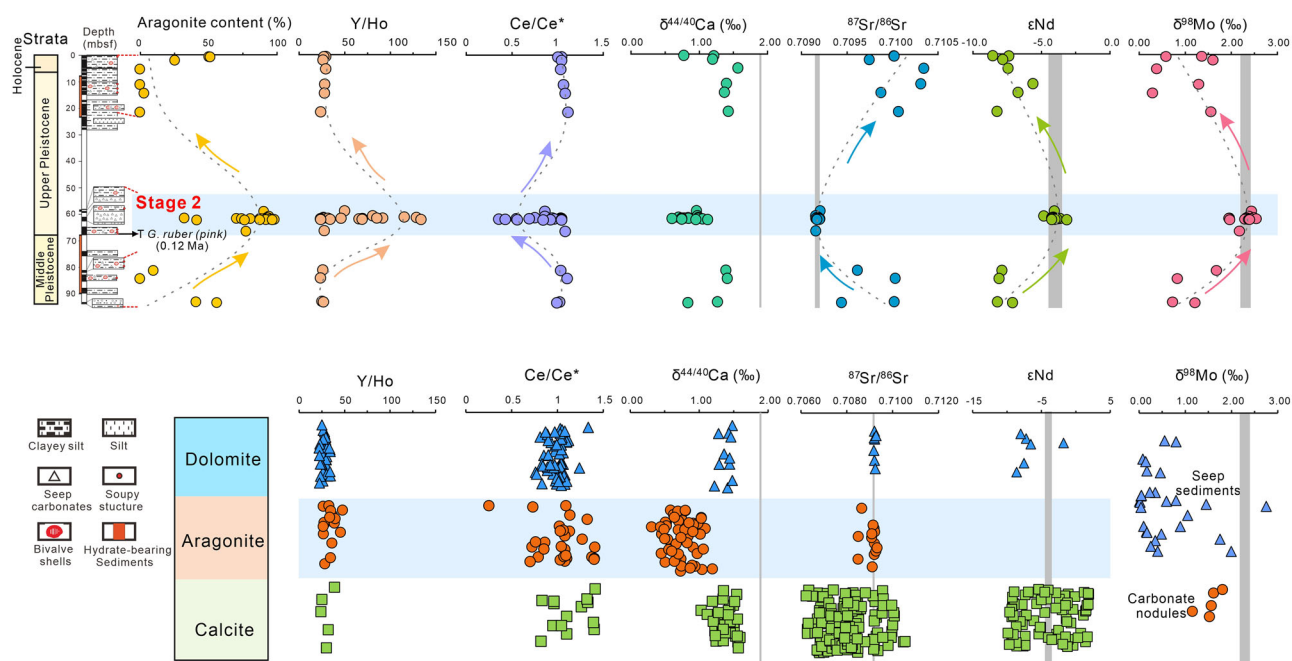


Fig. 2 | The vertical variations of aragonite content, Y/Ho, Ce/Ce*, $\delta^{98}\text{Mo}$, $^{87}\text{Sr}/^{86}\text{Sr}$, ϵNd , and $\delta^{44/40}\text{Ca}$ of seep carbonates at the GMGS2-08 site (top part) compared to those of seep carbonates and seep sediments from previous studies

(bottom part). The gray bars represent the estimated isotopic compositions of seawater. Errors of $\delta^{98}\text{Mo}$, $^{87}\text{Sr}/^{86}\text{Sr}$, ϵNd , and $\delta^{44/40}\text{Ca}$ values are less than the symbol size. The literature data are from refs. 19,24,25,30,31,34,37–39,75–77,86–92.

magnesium calcite (HMC), low-magnesium calcite (LMC), and dolomite, which were proven to be formed deeper in the sediment column with weak seepage and a low methane flux^{19,24}.

Results

Supplementary Tables 1 and 2 show the Mo, Ca, Sr, and Nd isotope data for seep carbonate from the GMGS2-08 site. The X-ray diffraction (XRD) results are listed in Supplementary Table 1. Nearly all of the carbonate cement in stage 2 is aragonite, whereas HMC (mean = 49%, $n = 25$), LMC (mean = 14%, $n = 25$), and dolomite (mean = 14%, $n = 25$) dominate carbonate crusts and microcrystalline matrix samples in stages 1 and 3 (Supplementary Table 1).

In general, the Mo, Ca, Sr and Nd isotopic values are similar between stages 1 and 3, which are greatly different from those of stage 2 (Fig. 2). As shown in Supplementary Table 2, the seep carbonates exhibit a large range of $\delta^{98}\text{Mo}$ values from 0.3 to 2.5‰ (mean = 1.8 ± 0.7 , $n = 28$) throughout the core. The $\delta^{98}\text{Mo}$ values of the seep carbonates range from 0.7 to 1.7‰ (mean = 1.1 ± 0.4 ‰, $n = 4$) in stage 1 and from 0.3 to 1.6‰ (mean = 1.0 ± 0.6 ‰, $n = 7$) in stage 3. In contrast, $\delta^{98}\text{Mo}$ values in stage 2 are higher, in the range of 1.9–2.5‰ (mean = 2.3 ± 0.2 ‰, $n = 17$).

The $\delta^{44/40}\text{Ca}$ values show a wide range, from 0.6 to 1.6‰ (mean = 1.0 ± 0.3 ‰, $n = 27$). The $\delta^{44/40}\text{Ca}$ values are positively correlated with Mg/Ca ratios ($r^2 = 0.47$, $n = 26$, $p < 0.0001$), but are negatively correlated with Sr/Ca ratios ($r^2 = 0.44$, $n = 26$, $p < 0.001$) (Fig. 3). The $\delta^{44/40}\text{Ca}$ values decrease from stage 1 (mean = 1.2 ± 0.3 ‰, $n = 4$) to stage 2 (mean = 0.9 ± 0.1 ‰, $n = 15$), followed by a rebound in stage 3 (mean = 1.3 ± 0.3 ‰, $n = 7$).

The $^{87}\text{Sr}/^{86}\text{Sr}$ ratios of carbonate samples vary between 0.709160 and 0.710336 (mean = 0.709472 ± 0.000415 , $n = 25$). The Sr isotopic ratios are also expressed in δ notation relative to that of the modern seawater $\delta^{87}\text{Sr}$ (‰) = $[(^{87}\text{Sr}/^{86}\text{Sr})_{\text{sample}} / (^{87}\text{Sr}/^{86}\text{Sr})_{\text{seawater}} - 1] \times 1000$, where $(^{87}\text{Sr}/^{86}\text{Sr})_{\text{seawater}}$ is ~ 0.709175 . Compared with carbonates in stage 2 ($\delta^{87}\text{Sr}$ varying from -0.021 to 0.049 , mean = 0.009 ± 0.024), carbonate $\delta^{87}\text{Sr}$ in stages 1 and 3 display a larger variation range from 0.049 to 1.637 , with a mean value of 0.937 ± 0.539 . Carbonate samples in stage 2 display a depletion in ^{143}Nd , with ϵNd values between -4.8 and -3.2 (mean = -4.0). Comparatively, the stages 1 and 3 samples exhibit lower ϵNd values from -8.5 to -7.6 (mean = -7.5) (Supplementary Table 2).

Discussion

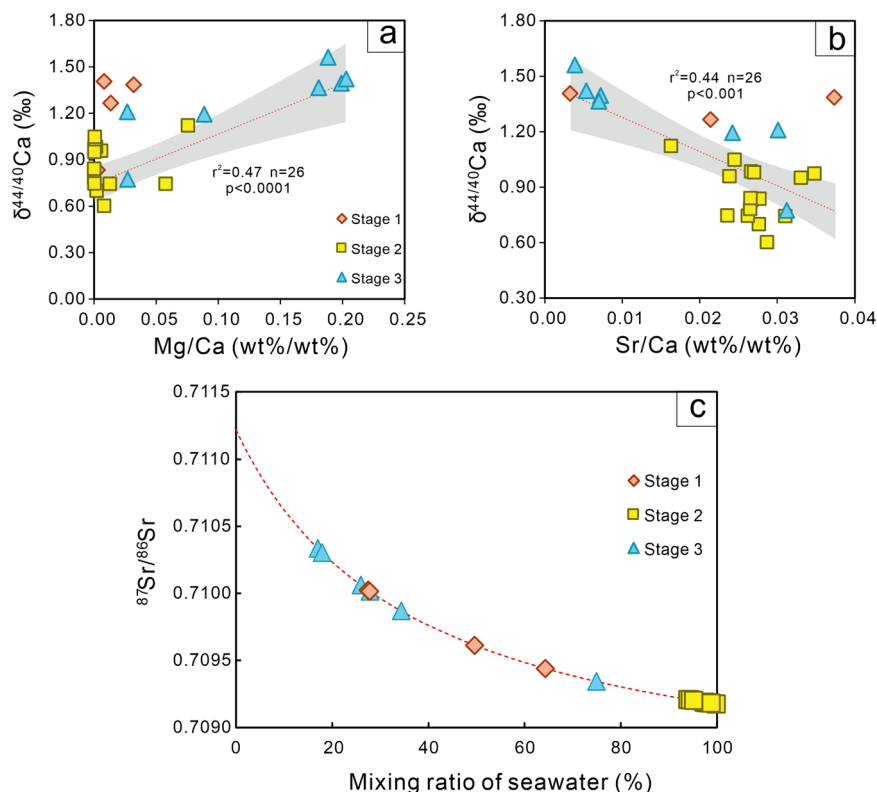
Calcium isotopes in seep carbonates: limitations as a proxy for ancient seawater

Aragonite precipitation typically occurs under high methane flux conditions with a shallow sulfate-methane transition zone (SMTZ) near the sediment-water interface, where sulfate is abundant, which can result in carbonate-cemented seafloor crusts³¹. However, dolomite and Mg-calcite (up to 20 mol % Mg) forms under lower methane flux when the SMTZ is within subsurface sediments and pore water sulfate is relatively low³². Therefore, the aragonite potentially preserves the seawater isotopic composition.

The $\delta^{44/40}\text{Ca}$ data for seep carbonates are 0.3–1.2‰ lower than that of modern seawater (~ 1.9 ‰)³³. Although carbonates in stage 2 are proven to be formed near the seafloor^{19,24}, their $\delta^{44/40}\text{Ca}$ values are even lower relative to those of seawater (Fig. 2). Thus, there is a range of isotope fractionations during the formation of seep carbonates, with the precipitated mineral typically enriched in ^{40}Ca relative to pore water or seawater. It is generally accepted that the Ca isotopic ratios in modern marine carbonate minerals are very different from one another. For instance, the Ca isotope fractionations between carbonate minerals and seawater can be anywhere from -1.8 to -0.8 ‰^{34,35}, which depends on the mineralogy and the rate of precipitation^{33,36}. However, investigations of Ca isotope fractionation during the formation of seep carbonate are scarce^{34,37–39}, particularly in a dynamic cold seep system with different seepage intensities.

Although the magnitude of Ca isotope fractionation during the precipitation of seep carbonate is less than that of modern marine carbonates³⁴, it is still a non-negligible fractionation^{37,38}. Seep carbonates are an important Ca sink in the cold seep sediments^{39,40}. Since ^{44}Ca is depleted in the near-surface region of carbonate minerals, rapid crystal growth leads to low $\delta^{44/40}\text{Ca}$ values in newly formed carbonate crystals⁴¹. Under thermodynamic disequilibrium conditions, Ca isotope fractionation in the growing calcite is primarily controlled by the preferential incorporation of light Ca isotopes⁴¹. Consequently, higher precipitation rates result in larger Ca isotope fractionation in the newly formed crystals^{41,42}. Controlled calcite precipitation experiments show that $\delta^{44/40}\text{Ca}$ decreases linearly with increasing growth rates⁴³. In a cold seep system, AOM produces bicarbonate, raising pore water pH and driving rapid carbonate precipitation. The rapid carbonate precipitation will cause ^{44}Ca enrichment in the pore water, because of the

Fig. 3 | The variations of $\delta^{44/40}\text{Ca}$ versus Mg/Ca and Sr/Ca for the seep carbonate and $^{87}\text{Sr}/^{86}\text{Sr}$ versus the mixing ratio of seawater. **a Relationship between $\delta^{44/40}\text{Ca}$ and Mg/Ca ratios in three-stage seep carbonates, with the red curve indicating the trend line. The gray area represents the 95% confidence interval. **b** Relationship between $\delta^{44/40}\text{Ca}$ and Sr/Ca ratios in three-stage seep carbonates, with the red curve indicating the trend line. The gray area represents the 95% confidence interval. **c** Relationship between $^{87}\text{Sr}/^{86}\text{Sr}$ and seawater mixing ratio in three-stage seep carbonates, with the red dashed line representing the mixing trend.**



preferred incorporation of ^{40}Ca into the precipitate through Rayleigh distillation^{37,39}. The precipitation rate is a primary factor controlling Ca isotope fractionation during the formation of seep carbonate³⁴. Aragonite forms at higher methane fluxes and precipitation rates compared to Mg-calcite³⁸, and the higher precipitation rates of seep carbonate during stage 2 lead to more negative Ca isotopes (Fig. 2). Aragonite $\delta^{44/40}\text{Ca}$ values diverge from seawater due to kinetic isotope effects during rapid carbonate precipitation under high methane flux. This change in precipitation rate is also supported by trace element distributions²⁴. $\delta^{44/40}\text{Ca}$ values are positively correlated with Mg/Ca ratios ($r^2 = 0.47$, $n = 26$, $p < 0.0001$), but are negatively correlated with Sr/Ca ratios ($r^2 = 0.44$, $n = 26$, $p < 0.001$) (Fig. 3), which are similar to previously published data of seep carbonates in the Barents Sea and the North Sea^{34,38}. Therefore, the Ca isotope in seep carbonate is not likely an ideal archive for the Ca isotopic composition of ancient seawater. While this complicates direct seawater $\delta^{44/40}\text{Ca}$ reconstruction, it provides a potential tracer for methane seepage intensity and carbonate precipitation rates.

Aragonite-dominated seep carbonates: reliable archives of seawater $\delta^{98}\text{Mo}$, ϵNd , and $^{87}\text{Sr}/^{86}\text{Sr}$

Molybdenum is a crucial element in biology, which may be attributed to its unique chemical properties, adaptation to higher availability in more oxygenated oceans through evolution, or association with sulfide minerals or organic matter during prebiotic chemical evolution in Mo-rich environments⁴⁴. Mo behaves conservatively in oxic seawater, where it is homogeneously distributed due to the long residence time of about 0.44–0.78 Ma^{5,45,46}. The anoxic sediments are major marine Mo sinks, and the Mo isotopic compositions of euxinic sediments can be used to trace paleo-ocean water Mo isotopic compositions if there is essentially complete scavenging of Mo from bottom waters with high concentration of hydrogen sulfide^{47,48}. The modern open-ocean seawater Mo reservoir is enriched in heavy isotopes ($2.3\text{‰} \pm 0.1\text{‰}$, 2SD, relative to NIST₃₁₃₄ = 0.25‰)⁴⁹.

The authigenic fraction of seepage carbonates is primarily composed of carbonate and sulfide minerals, as supported by petrographic evidence (Supplementary Fig. 1). Recent sequential extraction experiments on

authigenic carbonates from cold seeps in the Gulf of Mexico and SCS reveal that pyrite fractions sequester up to 90% of the total Mo inventory and dominate the Mo budget in these sediments⁵⁰. Minor fractions are associated with carbonate phases, organic matter, and Fe–Mn oxides⁵⁰. The Stage 2 aragonite-dominated seep carbonates formed near the sediment–seawater interface, preventing H_2S consumption during upward migration through the sedimentary column^{19,24}. The intense methane seepage may lead to episodic or continuous euxinic conditions (dissolved H_2S concentrations above $11\text{ }\mu\text{M}$) where H_2S was continuously formed in the bottom water during stage 2 by the AOM process^{19,24}. Building on prior constraints, our study reveals that Mo enrichment in aragonite-dominated carbonates reflects extreme sulfidation driven by sustained methane flux. Under these conditions, Mo deposition can be accelerated by one to two orders of magnitude, leading to minor or no isotopic fractionations due to the complete removal of tetrathiomolybdate from the euxinic water^{25,47}. This is evidenced by a positive correlation between $\delta^{98}\text{Mo}$ and Mo contents in the study area (Fig. 4a). Meanwhile, $\delta^{98}\text{Mo}$ also exhibit a positive correlation with the contents of aragonite ($r^2 = 0.64$, $n = 28$, $p < 0.0001$; Fig. 4b). The average value of $\delta^{98}\text{Mo}$ in aragonite-dominant seep carbonates (aragonite contents exceed 60%) in stage 2 is 2.3‰ , corresponding to those of modern seawater (Fig. 4b). Consequently, elevated Mo concentrations and $\delta^{98}\text{Mo}$ values in aragonite-dominated seep carbonates reflect intense sulfidation driven by methane flux. Under sulfidic conditions, Mo is scavenged as thiomolybdates (MoS_4^{2-}), incorporated into pyrite, and co-precipitated with carbonates (Detailed information of Mo isotope analysis can be found in the MATERIALS AND METHODS). Previous studies indicated that calcium carbonates formed in open marine environments only record the Mo isotopic composition of seawater when the sulfide concentration reaches a high level between 20 and $300\text{ }\mu\text{M}$ ⁵¹. Our study now documents seawater $\delta^{98}\text{Mo}$ values for aragonite-dominant seep carbonates that formed at times of intense methane fluid flow and episodic or continuous euxinic conditions with high enough free H_2S ($>>11\text{ }\mu\text{M}$) and dissolved Mo in the bottom waters²⁴. Thus, we demonstrate the potential of seep carbonates to preserve the seawater Mo isotopic signal, providing a novel proxy for reconstructing paleo-seawater chemical compositions.

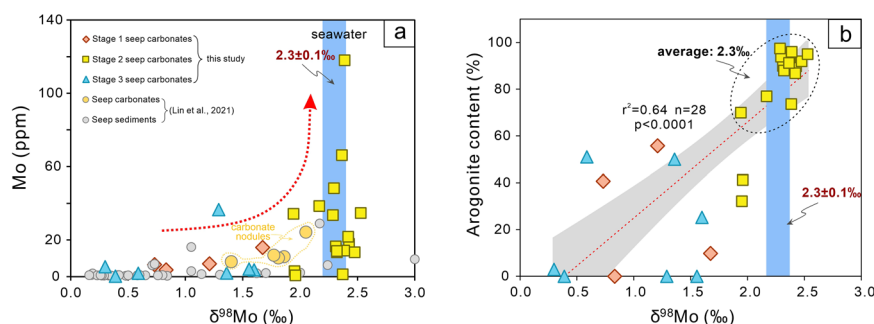


Fig. 4 | The variations of $\delta^{98}\text{Mo}$ versus Mo and aragonite content for the seep carbonates. **a Variations of Mo contents and $\delta^{98}\text{Mo}$ values of three-stage seep carbonates. The $\delta^{98}\text{Mo}$ values and Mo contents of seep sediments and carbonate nodules from ref. 25 are included for comparative analysis. The red curve indicates the variation trend. The blue bar represents the estimated $\delta^{98}\text{Mo}$ value of seawater.**

In marine environments, there is a variation in Mo isotope fractionation during the removal of Mo to sediments (Fig. 5). When black shale is formed in partially limited basins with euxinic bottom waters, it is likely to record the $\delta^{98}\text{Mo}$ signature of the global seawater⁴⁷. Hydrogenous Fe–Mn oxides are also a valuable tool to trace the evolution of seawater $\delta^{98}\text{Mo}$ ⁵². This approach exploits a consistent isotopic offset of $\sim 3\text{‰}$ between modern Mn oxides and seawater (Fig. 5b). In addition, seawater $\delta^{98}\text{Mo}$ can be recorded in some cases by primary carbonate precipitates and phosphorites⁵. However, a small Mo isotope fractionation is often observed, resulting in the wide range of $\delta^{98}\text{Mo}$ values in black shale, Fe–Mn oxides, carbonate and phosphorites (Fig. 5b). In fact, the content of hydrogen sulfide in water column is a key factor controlling the Mo isotopic compositions of sediments in reducing marine environments^{53,54}. Under conditions of low hydrogen sulfide content, the incomplete conversion of MoO_4^{2-} to MoS_4^{2-} would lead to Mo isotopic fractionation, resulting in a broad range of $\delta^{98}\text{Mo}$ values in sediments that do not accurately reflect the Mo isotopic composition of seawater^{53,55}. Additionally, if non-quantitative uptake occurs, the preferential incorporation of light Mo isotopes would lead to lower $\delta^{98}\text{Mo}$ values in seep sediments⁵³. In high-sulfide water columns, MoO_4^{2-} can be rapidly and efficiently converted into MoS_4^{2-} , enabling sulfide to effectively remove Mo from the water, thereby enriching Mo in the sediments while preserving the Mo isotopic composition of the surrounding water^{25,47}.

Furthermore, recent studies have highlighted the crucial role of the Fe–Mn particulate shuttle processes in Mo burial at high methane fluxes, identifying it as the primary factor responsible for the negative $\delta^{98}\text{Mo}$ values in seep sediments^{55,56}. This occurs because Fe–Mn oxyhydroxides tend to absorb lighter Mo isotopes from seawater and transfer this portion of Mo to seep sediments through reductive dissolution⁵⁷. Reductive dissolution of Fe–Mn particles at the sediment–water interface under anoxic and sulfidic conditions would release trace metals (e.g., Zn and Mo) into bottom water and pore water^{58,59}. The $\delta^{98}\text{Mo}$ values of carbonates in stages 1 and 3 are lower than that of seawater (Fig. 2), these values are similar to the previously published $\delta^{98}\text{Mo}$ data of seep carbonates in the SCS²⁵, which are most likely the result of the Fe–Mn particle shuttle processes. The $\delta^{98}\text{Mo}$ values of the aragonite-dominant seep carbonates (aragonite contents exceed 60%) are independent of Mo enrichment (Fig. 4b) and the contribution of oxyhydroxide-bound Mo is minimal, suggesting that oxyhydroxide have limited influence on $\delta^{98}\text{Mo}$ values of these carbonates. This further implies that Mo in the water column is nearly quantitatively removed to the seep carbonates. The Mo isotopic compositions (average: $\sim 2.30\text{‰}$) of aragonite-dominant seep carbonates may reflect the $\delta^{98}\text{Mo}$ value of bottom seawater, which is consistent with the values of modern seawater ($2.3\text{‰} \pm 0.1\text{‰}$)⁴⁹. A plausible explanation for the consistent isotopic composition between deep seawater and the aragonite-dominant seep carbonates is that the Mo contribution from Fe–Mn particle shuttling in the bottom water is minimal. Therefore, seep carbonate formed at a high level of hydrogen sulfide can effectively preserve the Mo isotopic signatures of contemporaneous

The yellow curve shows the range of Mo contents and $\delta^{98}\text{Mo}$ values in carbonate nodules. **b** Relationship between aragonite content and $\delta^{98}\text{Mo}$ values in three-stage seep carbonates. The red curve indicates the trend line. The blue bar represents the estimated $\delta^{98}\text{Mo}$ value of seawater. The gray area represents the 95% confidence interval.

seawater, and aragonite-dominant seep carbonates can serve as valuable records for reconstructing Mo isotopic variations in paleo-seawater.

Sr isotopes are considered to have a globally uniform distribution in seawater as the modern seawater residence time of Sr ($\sim 10^6$ years) is >1000 times longer than the ocean circulation time⁶⁰. Differences in Sr isotopic compositions in seep sediments are influenced by the compositions of seep fluids and the environmental conditions under which carbonate precipitation occurs^{61,62}. The Sr isotopic signatures of seep carbonates offer valuable insights into the processes of fluid–sediment interactions in cold seep environments^{61,63}. In methane seep areas, the seepage fluids, including methane, hydrogen sulfide, and other reducing species, mix with bottom seawater in varying proportions depending on methane seep flux, creating a dynamic geochemical environment^{24,64}. In areas of high-intensity methane seeps, aragonite typically forms directly from seawater^{24,64}. In such cases, the Sr isotopic compositions of the aragonite closely mirror that of seawater, with minimal fractionation (Fig. 2). However, in areas of low seep intensity, the chemical compositions of the seep fluids can modify the Sr isotopic signature of the carbonates, leading to potential deviations from typical seawater values⁶⁵. This occurs in cases where carbonates, such as dolomite and calcite, form under low methane seepage intensity⁶⁶, with the seep fluid contributing a greater proportion of Sr than the bottom seawater. Seep fluids are particularly enriched in radiogenic strontium isotopes and exhibit distinct isotopic signatures compared to ambient bottom seawater³⁰. In such scenarios, the Sr isotopic compositions of precipitated calcite or dolomite reflect a mixture of Sr isotopes from the seep fluid and the underlying seawater, rather than solely from the bottom seawater, leading to an isotopic signature that diverges from the typical oceanic Sr isotopic ratios.

In studies of cold seeps, Sr isotopic compositions have been utilized to determine the sources of fluids in both modern⁶¹ and ancient³⁰ seep environments. The contributions of seawater sources can be calculated following the methods of Viola et al.⁶⁵:

$$(^{87}\text{Sr}/^{86}\text{Sr})_{\text{sample}} = (^{87}\text{Sr}/^{86}\text{Sr})_{\text{sw}} * f_{\text{sr}} + (^{87}\text{Sr}/^{86}\text{Sr})_{\text{sf}} * (1 - f_{\text{sr}}) \quad (1)$$

$$f_{\text{sr}} = \text{Sr}_{\text{sw}} / (\text{Sr}_{\text{sw}} + \text{Sr}_{\text{sf}}) \quad (2)$$

where sw means seawater, and sf is seep fluid. Sr_{sw} and Sr_{sf} are Sr concentrations in seawater and seep fluid (mg/L), respectively. Sr concentration of seawater (8.10 mg/L) and seep fluid (2.179 mg/L) in the SCS were estimated by Yu et al.⁶⁷. Modern seawater has a $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.709175^{30,68}, and seep fluid has a $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.711219³⁰. Calculated seawater fractions vary between 17% and 100% (Fig. 3c). The aragonite-dominant seep carbonates in stage 2 show a major contribution from seawater (94%–100%), whereas the HMC, LMC and dolomite in stages 1 and 3 show a mix of 6%–83% seep fluid and 17%–94% seawater. $^{87}\text{Sr}/^{86}\text{Sr}$ ratios in seep carbonates with a typical coeval seawater value were also reported from the

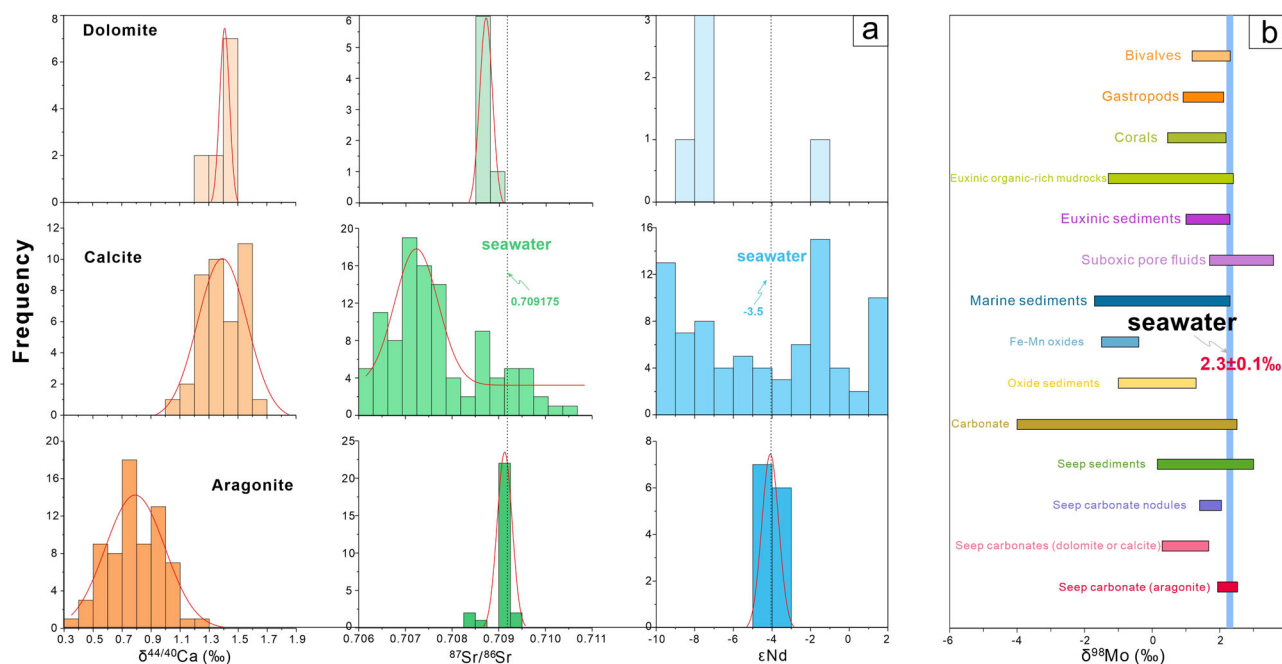


Fig. 5 | Histogram of $\delta^{44/40}\text{Ca}$, $^{87}\text{Sr}/^{86}\text{Sr}$, and ϵNd of the seep carbonates (dolomite-, calcite-, aragonite-dominated) and $\delta^{98}\text{Mo}$ values of different biological and marine sediment samples. **a Histograms of $\delta^{44/40}\text{Ca}$, $^{87}\text{Sr}/^{86}\text{Sr}$, and ϵNd values in (dolomite-, calcite-, and aragonite-dominated) seep carbonates. Orange, green, and blue bars represent $\delta^{44/40}\text{Ca}$, $^{87}\text{Sr}/^{86}\text{Sr}$, and ϵNd values, respectively. Black dashed lines**

indicate seawater $^{87}\text{Sr}/^{86}\text{Sr}$ and ϵNd values. **b** Comparative $\delta^{98}\text{Mo}$ values of different biological and marine sediment samples. The blue bar represents the estimated $\delta^{98}\text{Mo}$ value of seawater. The isotope data are from refs. 19,24,25,30,31,34,37–39,75–77,86–95.

North Sea³², the Cascadia margin⁶³, the Gulf of Mexico⁶², the Gulf of Cadiz⁶⁹, the mid-Atlantic Margin⁷⁰, and the SCS³⁰. In summary, the differences among cold seep carbonates highlight the critical role of seep fluid compositions and seepage intensity in determining the Sr isotopic signatures preserved in these sediments. Aragonite-dominated seep carbonates, which form in environments with intense methane seepage, are more likely to accurately record the Sr isotopic composition of the surrounding seawater.

Neodymium isotopic signatures of authigenic sedimentary phases were also widely used for reconstructing past ocean circulation patterns^{7,71}. The use of Nd isotopes as a seawater circulation proxy is grounded in its quasi-conservative behavior in seawater and the geographically differences in ϵNd value based on water mass origin⁷². However, recent studies have shown that lithogenically sourced Nd could impact the Nd isotopic signature of the authigenic component, with the authigenic record of sediments being heavily affected by detrital components, potentially leading to the misrepresentation of bottom water ϵNd values⁶. Theoretically, aragonite precipitated under high methane seepage intensity has the potential to more faithfully preserve the Nd isotopic characteristics of seawater. In these high-flux environments, where rapid precipitation occurs in direct contact with seawater, aragonite can capture a Nd isotopic signature that closely mirrors the surrounding water mass^{7,73,74}. Instead, dolomite and calcite that form in pore water environments are more influenced by the Nd isotopic signature of pore water from which they precipitate^{30,75–77}. Neodymium isotopic signatures of pore water are typically governed by post-deposition processes (such as mineral dissolution, organic matter degradation, and adsorption/desorption reactions), leading to an Nd isotopic composition that is distinct from that of bottom seawater^{7,74}.

The spatial distribution of seawater ϵNd in the northern SCS has shown that ϵNd of the deep-water masses of the SCS below 1500 m is homogenous (−3.5 to −4.5) and displays similar ϵNd values to those of the Pacific Deep Water (−3.9 to −4.4)^{78,79}. This implies that the ϵNd values of the Pacific Deep Water, which enters the northern SCS through the Luzon Strait, are not modified by the local process⁷⁹. This contrasts with the southern SCS, where ϵNd is slightly more radiogenic due to greater influence of local

riverine inputs (e.g., Pearl River: −10; Mekong River: −8)⁷⁹. The ϵNd levels in stages 1 and 3 were between −8.5 and −5.6 (mean = −7.5), which is similar to previously published ϵNd data of seep carbonates^{30,75–77} (Fig. 2). These results are different from those of the seawater (−3.5 to −4.5)^{78,79} and bulk sediments (−11.7 to −8.8)^{78,80} in the SCS. The precipitation of HMC, LMC, and dolomite took place within sediment pore water⁶¹, and thus ϵNd values of seep carbonate in stages 1 and 3 are similar to those of the Fe–Mn oxyhydroxides (−8.2 to −6.2)⁸¹ and organic matters (−6.7)⁸². This supports the previous suggestion that pore water ϵNd is controlled mainly by authigenic components, i.e., organic matter and Fe–Mn (oxyhydr) oxides⁷. Conversely, the average value of ϵNd in the aragonite-dominant seep carbonates in stage 2 is −4.1 (Supplementary Table 2), which is about the same as the value of seawater in SCS (−3.5 to −4.5)^{78,79}. This Pacific-like signature contrasts with radiogenic pore fluids (−8 to −11) and aligns with mid-Holocene coral records (−4.2), confirming that aragonite preserves seawater ϵNd ⁷⁹. Therefore, the Nd isotopic composition of dolomite- and calcite-dominant seep carbonates formed in low methane seepage environments tends to reflect local sedimentary influences on pore water, causing their isotopic signatures to differ from bottom seawater (Fig. 2). Notably, aragonite-dominant seep carbonates can effectively record the Nd isotopic information of the bottom seawater, largely because aragonite tends to precipitate rapidly from seawater in high methane seepage environments³⁸. The rapid precipitation of aragonite at the sediment–seawater interface limits the influence by pore waters, thus preserving the Nd isotope signature of the seawater at the time of precipitation³⁸. These results indicate that the Nd isotopic signal in aragonite has not been blurred by diagenetic alteration, and thus can record the Nd isotopic signal of seawater. The rapid precipitation of aragonite-dominated carbonates in closed systems could preserve the original fluid signatures, while their high Sr/Ca ratios and resistance to diagenetic alteration ensure reliable isotopic records. Consequently, aragonite-dominated carbonates offer distinct advantages over traditional proxies like biogenic shells for reconstructing seawater Sr–Nd isotopes.

To date, our study now documents the values of $\delta^{98}\text{Mo}$, ϵNd , and $^{87}\text{Sr}/^{86}\text{Sr}$ for aragonite-dominant seep carbonates that formed at the SMTZ

close to the seafloor, which show no fractionation relative to those of seawater. However, variations in mineralogy and precipitation rate are most likely what led to the existence of Ca isotope fractionation. Still, our study indicates the potential of seep carbonates to preserve the isotopic compositions of seawater. Further studies on other aragonite in seep carbonates are required to determine whether seep carbonates formed during periods of intense fluid flow can firmly record the isotopic composition of the coeval seawater as observed in this study.

Materials and methods

Mineralogy analysis

XRD is used for semi-quantitative carbonate mineralogy via a DXR 3000 diffractometer (Rigaku, Japan) equipped with a diffracted beam graphite monochromator and using CuK α radiation at the Guangzhou Marine Geological Survey (GMGS, Guangzhou, China). Scans were done from 5° to 60° (2 θ) at 0.02°/s, using 40 mA current and 40 kV acceleration voltage. Rietveld analysis of the diffractograms with the program SIROQUANT was used to figure out the relative amounts of carbonate minerals. LMC represents calcite with MgCO₃ less than 5 mol%, HMC stands for calcite with 5–20 mol% MgCO₃, and dolomite is carbonate phases with 30–50 mol % MgCO₃.

Mo isotope analysis

Analysis of Mo isotopes was performed on a multicollector inductively coupled plasma mass-spectrometer (MC-ICP-MS, Thermo-Fisher Scientific Neptune) in the State Key Laboratory of Geological Processes and Mineral Resources (GPMR), China University of Geosciences in Wuhan, using a double isotope tracer (¹⁰⁰Mo, ⁹⁷Mo) for correction of instrumental mass bias. The authigenic fractions of seep carbonates include carbonate and sulfide minerals, which (particularly pyrite) account for up to 90% of the Mo budget, rendering the influence of the detrital fraction quite limited. Consequently, for cold seep carbonates, analyzing the Mo isotopes of the carbonate fraction alone holds little significance. Therefore, in this study, we measured the Mo isotopic composition of whole-rock samples, which effectively represents the Mo isotopic composition of the authigenic components within the cold seep carbonates. Samples were digested in a mixture of HNO₃, HClO₄, and HF. Prior to analyses, Mo was extracted and purified by using a custom-made extraction chromatographic resin of N-benzoyl-N-phenyl hydroxylamine⁸³. All procedures were carried out in a clean laboratory environment. Isotope ratios of Mo are expressed as $\delta^{98}\text{Mo}$ relative to NIST SRM 3134 ($\delta^{98}\text{Mo}_{\text{NIST3134}} = 0.25\text{‰}$). Reference material (GBW07316 marine sediment) was also repeatedly measured along with the samples. The $\delta^{98}\text{Mo}$ values of NIST SRM 3134 and GBW07316 were determined to be $0.25 \pm 0.05\text{‰}$ (2SD, $n = 18$), and $-0.46 \pm 0.06\text{‰}$ (2SD, $n = 3$), which is in excellent agreement with previously published values⁸⁴. The results in the conventional δ -notation in (‰) are equivalent to milli Urey (mUr).

Sr and Nd isotope analysis

Sr and Nd isotopic ratios were analyzed following the method of Ge et al.³⁰ at the GPMR, using a Finnigan Triton TIMS. Approximately 10–20 ml of the acetic acid leachate stock solution was used to rinse the carbonate components from seep carbonates. This ensures the accurate acquisition of the Sr–Nd isotopic compositions of the authigenic carbonate components in the seep carbonates, free from the influence of Sr–Nd derived from detrital sources. The samples were then purified in a clean laboratory using ion exchange columns of Dowex AG50WX12 cation resin and Eichrom Ln-Spec resin successively to separate Sr and Nd. Isotopic ratios of ¹⁴³Nd/¹⁴⁴Nd and ⁸⁷Sr/⁸⁶Sr were normalized to ¹⁴⁶Nd/¹⁴⁴Nd = 0.721900 and ⁸⁸Sr/⁸⁶Sr = 8.375209, respectively. Measurements of standard JNdi-1 and SRM NBS987 give average values of 0.512118 ± 0.000007 ($n = 25$) and 0.710254 ± 0.000008 ($n = 22$) for ratios of ¹⁴³Nd/¹⁴⁴Nd and ⁸⁷Sr/⁸⁶Sr, respectively. Nd isotopic analyses are reported in the standard epsilon notation $\epsilon\text{Nd}(t) = [({}^{143}\text{Nd}/{}^{144}\text{Nd})_{\text{sample}(t)} / ({}^{143}\text{Nd}/{}^{144}\text{Nd})_{\text{CHUR}(t)} - 1]10^4$, where CHUR is the chondritic uniform reservoir with a present day ¹⁴³Nd/¹⁴⁴Nd = 0.512638.

Ca isotope analysis

The TIMS method described by Feng et al.⁸⁵ was used to do the Ca isotope analyses in the GPMR with a ⁴²Ca–⁴⁸Ca double spike. In brief, sample powder was weighted into a homemade PTFE-lined steel bomb, and wetted with 0.5 ml of H₂O. After the addition of 1 ml HNO₃ and 1 ml HF, samples were maintained at 190 °C for 48 h for digestion. The contents were dried and redissolved with 1 ml of HNO₃ twice to completely remove HF. Finally, the solution was evaporated to dryness, redissolved in 1 ml of purified HNO₃ and evaporated to dryness again. Certain amounts of ⁴²Ca–⁴⁸Ca double spike were added to optimize the test precision. Then the mixture, including the sample and double spike, was purified from matrix elements prior to mass spectrometry using DGA resin (Eichrom, Bruz, France) in gravity flow micro-columns. The purified Ca was then loaded with 20% H₃PO₄ (Merk KGaA, Darmstadt, Germany) on Re double filaments and introduced into the TIMS for mass spectrometric analysis. Referring the measured values to the international standards, NIST SRM915a as a common standard allows for the comparison of data from different laboratories. The results of the Ca isotope expressed in delta (δ) notation $\delta^{44/40}\text{Ca} = [(\delta^{44/40}\text{Ca})_{\text{sample}} / (\delta^{44/40}\text{Ca})_{\text{std}} - 1]10^3$. Std refers to the NIST SRM915a, and the sample donates data corrected for instrumental fractionation. The analytical precision achieved by this procedure is 0.14‰ (2 σ , at the 95% confidence level). The total Ca blank was typically less than 10 ng.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All relevant data to reproduce the manuscript findings can be found online at <https://doi.org/10.5281/zenodo.15124252>.

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Author contributions

Yinan Deng: conceptualization, investigation, funding acquisition, data curation, data interpretation, writing—original draft, writing—reviewing & editing. Qingjun Guo: conceptualization, investigation, writing—reviewing & editing. Ganglan Zhang: visualization, writing—reviewing & editing. Fang Chen: writing—reviewing & editing. Shengxiong Yang: funding acquisition, writing—reviewing & editing. Gaowen He: writing—reviewing & editing. Hongfei Lai: writing—reviewing & editing. Daohua Chen: writing—reviewing & editing. Yunxin Fang: funding acquisition. Zenggui Kuang: funding acquisition. Jun Cao: writing—reviewing & editing. Yangtao Zhu: methodology, writing—reviewing & editing. Dawuda Usman: writing—reviewing & editing. Yufei Liu: methodology, writing—reviewing & editing. Bin Zhao: writing—reviewing & editing. Xuexiao Jiang: methodology. Mingyu Zhao: writing—reviewing & editing.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Yinan Deng or Qingjun Guo.

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